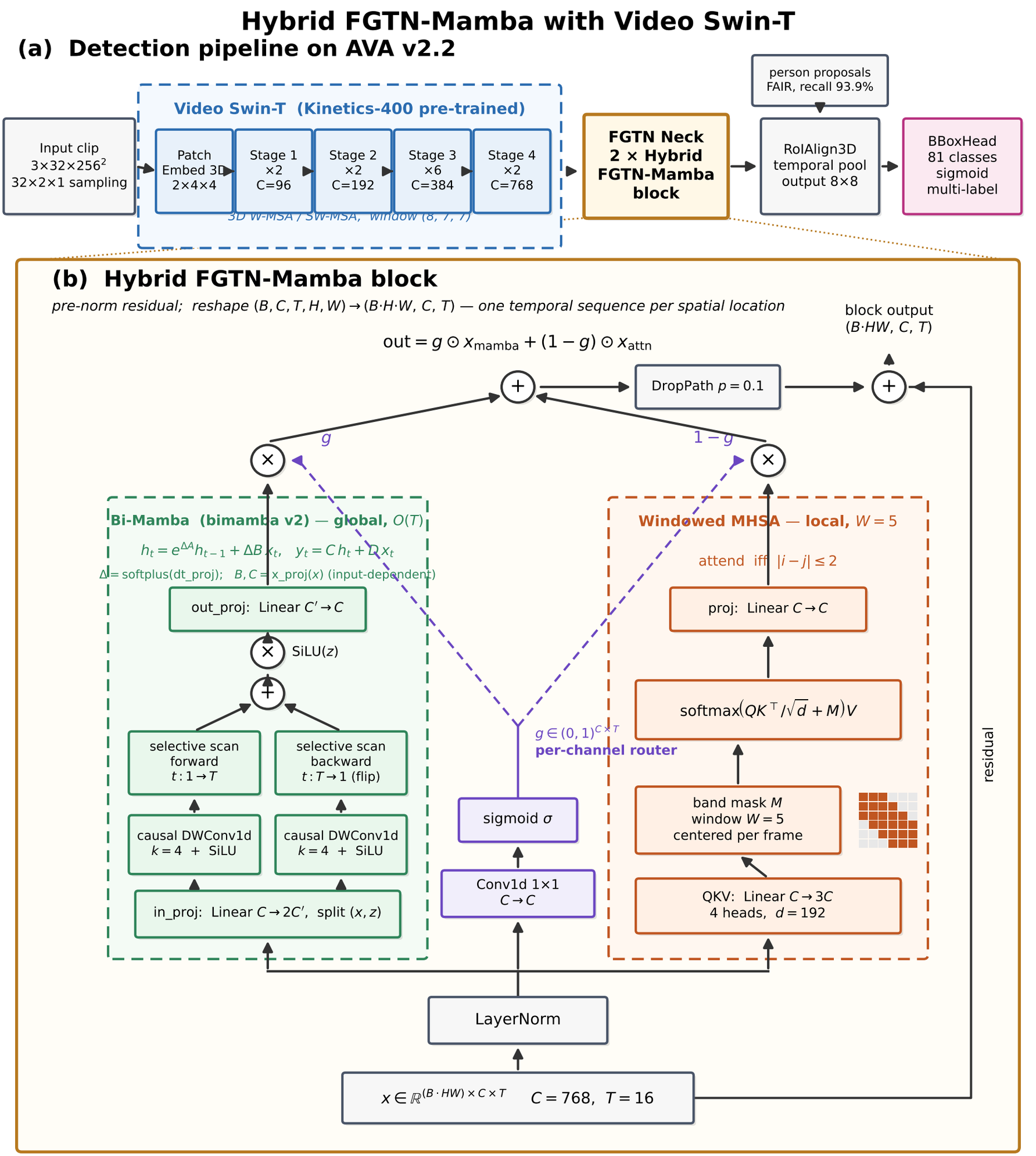

Hybrid FGTN-Mamba with Video Swin-T
(a) Detection pipeline on AVA v2.2
Input clip
3×32×256²
32×2×1 sampling
Video Swin-T (Kinetics-400 pre-trained)
Patch Embed 3D
2×4×4
Stage 1
×2
C=96
Stage 2
×2
C=192
Stage 3
×6
C=384
Stage 4
×2
C=768
3D W-MSA / SW-MSA, window (8, 7, 7)
FGTN Neck
2 × Hybrid
FGTN-Mamba
block
person proposals
FAIR, recall 93.9%
RoIAlign3D
temporal pool
output 8×8
BBoxHead
81 classes
sigmoid
multi-label
(b) Hybrid FGTN-Mamba block
pre-norm residual; reshape (B, C, T, H, W) → (B·H·W, C, T) — one temporal sequence per spatial location
block output
(B·HW, C, T)
out = g ⊙ x_mamba + (1 − g) ⊙ x_attn
DropPath p = 0.1
g
1 − g
Bi-Mamba (bimamba v2) — global, O(T)
h_t = e^{ΔA} h_{t−1} + ΔB x_t, y_t = C h_t + D x_t
Δ = softplus(dt_proj); B, C = x_proj(x) (input-dependent)
out_proj: Linear C′ → C
SiLU(z)
selective scan
forward
t : 1 → T
selective scan
backward
t : T → 1 (flip)
causal DWConv1d
k = 4 + SiLU
causal DWConv1d
k = 4 + SiLU
in_proj: Linear C → 2C′, split (x, z)
Windowed MHSA — local, W = 5
attend iff |i − j| ≤ 2
proj: Linear C → C
softmax(QK^⊤/√d + M)V
band mask M
window W = 5
centered per frame
QKV: Linear C → 3C
4 heads, d = 192
g ∈ (0, 1)^{C×T}
per-channel router
sigmoid σ
Conv1d 1×1
C → C
LayerNorm
x ∈ ℝ^{(B·HW)×C×T} C = 768, T = 16
residual